

# Seoyeon Choi

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## Education

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**Ph.D. in Mechanical Engineering**, University of California, Berkeley Aug 2024 - Present

- Advisor: Prof. Negar Mehr | Research Area: Robotics, AI, and Control
- Affiliated in Berkeley Artificial Intelligence Research (BAIR) Lab

**B.S. in Mechanical Engineering**, Yonsei University Mar 2019 - Aug 2023

- GPA: 3.95/4.0, *Summa Cum Laude*
- Senior Thesis: Reinforcement learning for force control of robot manipulator in contact-rich tasks

## Research Interest

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Robot Learning, LLMs and VLMs for Robotics, Humanoid, Multi-agent systems, Reinforcement Learning

## Selected Publications and Conference

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See full list: [Google Scholar](#) [🔗](#)

\*equal contribution

- **S. Choi**, S. Ye, N. Bui, A. Shrivastava, K. Ryu, D. Tirumala, M. Wulfmeier, and N. Mehr, "HuGo: LLMs as Whole-Body Policy Code Designers for Humanoid Loco-Manipulation", Preprint ([Link](#) [🔗](#))
- H. Park\*, J. Chae\*, M. Park\*, S. Park, H. Yoo, **S. Choi**, S. Yoo, J. Seo, S. Idrees, J. Hyun, J. Lee, R. Horowitz, Y. Lee, and J. Choi, "CoHuB: A Simulation Benchmark for Multi-Humanoid Collaboration", Preprint ([Link](#) [🔗](#))
- **S. Choi**\*, K. Ryu\*, J. Ock, and N. Mehr, "CRAFT: Coaching Reinforcement Learning Autonomously using Foundation Models for Multi-Robot Coordination Tasks", Preprint, **Outstanding Paper Award in ICRA 2026 Workshop** on Scaling Compositional Intelligence for Multi-Agent Robotic Systems ([Link](#) [🔗](#))
- J. Seo, A. Kruthiventy, S. Lee, M. Teng, X. Zhang, **S. Choi**, J. Choi, and R. Horowitz, "EquiContact: A Hierarchical SE (3) Vision-to-Force Equivariant Policy for Spatially Generalizable Contact-rich Tasks.", Preprint ([Link](#) [🔗](#))
- D. Dong\*, M. Bhatt\*, **S. Choi**, and N. Mehr, "MIMIC-D: Multi-modal Imitation for Multi-agent Coordination with Decentralized Diffusion Policies", **2026 IEEE International Conference on Robotics and Automation (ICRA)** ([Link](#) [🔗](#))

## Research Experience

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**Intelligent Control (ICON) Lab**, University of California, Berkeley Aug 2024 - Present

Graduate Assistant, *Advisor*: Prof. Negar Mehr

### Sim-to-Real Adaptation of Humanoid World Models

- Extending [SimDist](#) [🔗](#) world-model adaptation to humanoids, which face much larger sim-to-real gaps.

### Humanoid Loco-Manipulation using LLM-Generated Policy Code

- Proposed *HuGo*, a framework in which LLMs write whole-body policy code for humanoid loco-manipulation from task descriptions alone, without demonstrations or reward engineering.
- Designed an iterative refinement loop where LLMs analyze rollouts and apply localized code edits, improving real-world success on Unitree G1 from 20% to 90% without retraining.

### Learning Multi-robot Coordination using Foundation Models

- Proposed *CRAFT*, a curriculum-learning framework in which foundation models (LLMs/VLMs) act as coaches for multi-robot coordination tasks.

- Designed a closed-loop pipeline where foundation models generate curricula, refine rewards, and evaluate rollouts to improve policy learning.

### Multi-agent Imitation Learning using Diffusion Policies

- Proposed *MIMIC-D*, a decentralized multi-agent imitation learning framework that captures multimodality in expert demonstrations using diffusion policies.
- Demonstrated MIMIC-D on decentralized bimanipulation task, using only local observations for each agent.

### Machine Learning and Control Systems (MLCS) Lab, Yonsei University

Jan 2022 - Jul 2024

Undergraduate Intern, *Advisor*: Prof. Jongeun Choi

- Developed a visuomotor policy for contact-rich task that learns end-effector forces and gains of admittance impedance controller using reinforcement learning

## Project Experience

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### Development of Driver Models utilizing Reinforcement Learning, Hyundai Mobis

Aug 2023 - Feb 2024

- Developed a reinforcement learning algorithm to optimize parameterized cost function scheduling in model predictive control for rear-wheel-steer model-in-loop simulation evaluation system.

### ResQ4U: Unmanned Life-saving Rescue Launcher Robot, Yonsei University

Mar 2023 - Jun 2023

- Created and designed a rescue robot that detects and launches a life jacket at a drowning person.
- Developed a mechanism to detect the person with an object detection machine learning model using an IR-RGB camera and a one-point LiDAR, and designed the actuators of pitching and aiming mechanism.

### 2022 Autonomous Driving Robot Racing Contest, Korea Robotics Society

Jan 2022 - Nov 2022

- Developed a control algorithm utilizing LiDAR, IMU, and GPS for collision avoidance with other vehicles.
- Implemented Dijkstra and Regulated Pure Pursuit for planning and extended Kalman filter for localization.

## Skills

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**Programming:** C/C++, Python, MATLAB/Simulink

**Software & Tools:** PyTorch, JAX, TensorFlow, Mujoco, IsaacGym/Sim, ROS, CAD, 3D Printing

**Hardware:** Jetson, Raspberry Pi, Arduino, Motion Capture (OptiTrack, Vicon)

**Robots:** Humanoid (Unitree G1), Manipulator (Panda, Kinova, xArm), Quadruped (Unitree Go1, Go2), Turtlebot3, Scout Mini

## Selected Honors, Awards and Services

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### Honors

Summa Cum Laude (Top 1% in College of Engineering), Yonsei University

Aug 2023

### Scholarships

National Science and Technology Scholarship, National Research Foundation of Korea  
Yonsei Eminence Scholarship, Yonsei University

Mar 2021 - Aug 2023  
Sep 2019 - Feb 2021

### Organizer

Head Student Organizer, Bay Area Robotics Symposium (BARS)

2026

### Reviewer

IEEE Robotics and Automation Letters (RA-L)

2026, 2025

IEEE International Conference on Robotics and Automation (ICRA)

2026, 2025

Conference on Robot Learning (CoRL)

2026